SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

SOFTWARE REQUIREMENTS SPECIFICATION

Digital Twin of a University Building for
Occupancy-Aware Energy and Indoor Environmental Monitoring
CampusTwin EnergyMind

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Source basis: Product Requirements Document (PRD) and Phase-I Major Project Report. Where the two sources differ, the discrepancy is explicitly identified rather than silently resolved.

Document Control

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Primary Sources	PRD; Phase-I Major Project Report
Implementation Boundary	Monitoring and decision-support prototype
Current MVP Mapping	Three IILM University buildings specified in this SRS
Unresolved Scope Item	2D MVP vs 3D Digital Twin direction — TBD

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification defines the functional, non-functional, data, security, interface, validation, and acceptance requirements for CampusTwin EnergyMind. The SRS translates the project PRD and Phase-I scope into clear, testable requirements for development and academic evaluation.

1.2 Project Scope

CampusTwin EnergyMind is a web-based Digital Twin prototype for monitoring room occupancy, indoor environmental conditions, energy consumption, HVAC status, and lighting status. The system associates room-level telemetry with a digital spatial representation so authorized users can understand current building

conditions, review historical trends, detect energy-waste situations, and receive decision-support recommendations.

The current MVP mapping in this SRS covers three IILM University buildings: SVH, Engineering Block, and Law Block. Although the PRD describes a one-building MVP, the project team has supplied the three-building mapping for the current SRS baseline. The architecture shall remain extensible for additional buildings.

1.3 Intended Audience

Audience	Use of this SRS
Project team / developers	Implementation reference and requirement baseline.
Faculty / project guide	Evaluation of scope, completeness, feasibility, and testability.
Testing / validation team	Derivation of test cases and acceptance checks.
University facility stakeholders	Understanding system capabilities and limitations.
Students / faculty users	Understanding read-only room visibility where applicable.

1.4 Product Overview

The product combines a spatial Digital Twin interface with telemetry ingestion, data processing, PostgreSQL storage, historical analytics, rule-based energy-waste detection, alerts, and recommendations. The initial implementation may use simulated data; ESP32 and MQTT can be used for selected real sensor streams where hardware, network access, and permissions are available.

1.5 Definitions and Acronyms

Term	Meaning
Digital Twin	Digital representation of building spaces associated with current or historical data.
MVP	Minimum Viable Product / current prototype scope.
Telemetry	Time-stamped measurement or state data received from a source.
RBAC	Role-Based Access Control.
HVAC	Heating, Ventilation and Air Conditioning.
MQTT	Message Queuing Telemetry Transport.
PII	Personally Identifiable Information.
kW / kWh	Active power / energy consumed.
TBD	To Be Determined / requiring project-team confirmation.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The product sits between physical/simulated building data and human decision-making. Data sources provide occupancy, environmental, energy, and resource-state information. The backend validates and processes the data, stores historical records, synchronizes room states with the Digital Twin, and exposes information to the React web interface. The system is not a replacement for an existing Building Management System and does not directly actuate physical infrastructure.

2.2 Product Functions

- Authenticate users and apply role-based access.
- Represent the three current MVP buildings, their mapped floors, and room IDs.
- Display occupancy, temperature, humidity, CO_2, energy, HVAC, and lighting information where valid data exists.
- Detect configured occupancy-aware energy-waste conditions and generate alerts.
- Provide current-state dashboards and historical occupancy-versus-energy analytics.

- Provide a configurable simulator for demonstration and testing.
- Provide recommendations as human decision-support outputs.
- Support ingestion from APIs and, where available, ESP32/MQTT sensor streams.

2.3 User Classes

User Class	Primary Needs
University Facility / Building Administrator	Building-wide visibility, utilization trends, major anomalies, administrative controls.
Energy / Resource Manager	Occupancy-energy correlation, waste alerts, historical energy analysis.
Campus Operations Staff	Operational waste locations and inspection-oriented alerts.
Faculty / Students	Read-only room availability and current occupancy visibility.

2.4 Operating Environment

Layer	Environment / Requirement
Client	Modern desktop web browser; Chrome, Edge, or Firefox.
Display	Responsive desktop layouts including 1920×1080 and 1366×768.
Frontend	React.js; Three.js / React Three Fiber where the confirmed Digital Twin direction requires it.
Backend	Python / FastAPI.
Database	PostgreSQL.
IoT	ESP32 and MQTT for selected real telemetry, subject to availability.
Analysis	Python / Scikit-learn for analysis or future occupancy prediction.

2.5 Constraints

- Web-only deployment target for the MVP; native mobile applications are out of scope.
- Real sensor deployment depends on hardware, network infrastructure, campus access, and permissions.
- Floorplan assets may be simplified SVG/JSON layouts for MVP implementation.
- The system is monitoring/decision support only; it shall not autonomously control HVAC, lighting, breakers, or other critical infrastructure.
- Privacy scope is limited to aggregate occupancy information; no facial recognition, biometrics, or individual identity tracking is required.

2.6 Assumptions and Dependencies

- Synthetic data can provide complete test coverage if real telemetry is unavailable.
- Simulated and real telemetry should use compatible data structures so that one can replace the other without redesigning the complete application.
- Digital floor-layout assets are available or can be prepared in a simplified representation.
- Operational building-hour availability is treated as 07:00–22:00 for the documented availability target.
- Exact environmental thresholds are configurable/TBD unless explicitly defined in project documentation.

2.7 MVP Scope

Building	Mapped Floors / Room IDs
SVH — Sir Vishveshwarya Hall	Ground: 101–112; Second: 201–213
Engineering Block	Ground: 101–108; First: 201–208; Second: 301–308; Third: 401–408
Law Block	Ground: 101–108; First: 201–208; Second: 301–308; Third: 401–408

Room IDs are preserved exactly as supplied. Where room numbering and physical floor naming require clarification, the physical mapping is treated as TBD rather than being inferred.

2.8 Out-of-Scope Features

- Autonomous switching/control of physical HVAC, lighting, breakers, valves, or other critical infrastructure.
- Facial recognition, biometric monitoring, or individual student/staff tracking.
- Native iOS/Android applications.
- VR/AR spatial walkthroughs.
- Campus-wide federation as part of the current MVP.
- Advanced predictive maintenance and forecasting beyond the agreed analysis/prediction scope.

3. USER ROLES AND PERMISSIONS

The permission matrix below separates confirmed project interactions from permissions that require implementation detail. Any permission marked Proposed is not treated as a final source-of-truth requirement.

Feature	Facility Admin	Energy Manager	Operations Staff	Faculty / Student
Login / logout	Yes	Yes	Yes	Yes, if enabled
View building/floor/room status	Yes	Yes	Yes	Read-only
View occupancy	Yes	Yes	Yes	Read-only
View environmental metrics	Yes	Yes	Yes	Proposed read-only
View energy/resource state	Yes	Yes	Yes	Proposed read-only
View alerts	Yes	Yes	Yes	No / Proposed
Acknowledge operational alert	Proposed	Proposed	Proposed	No
Configure waste-rule thresholds	Proposed	Proposed	No	No
Run/control simulator	Yes / Proposed	Yes / Proposed	Proposed	No
Historical analytics	Yes	Yes	Yes / Proposed	No / Proposed
Export summaries	Yes	Yes	Proposed	No
User/role administration	Proposed	No	No	No

4. SYSTEM FEATURES

Module	Purpose	Primary Users
Authentication	Secure sign-in and access control.	All authenticated users
Digital Twin Visualization	Spatial representation of buildings, floors and rooms.	All; role-dependent detail
Floor and Room Navigation	Navigate mapped floors and select rooms.	All
Occupancy Monitoring	Show occupied/unoccupied/available status and counts where available.	All
Indoor Environmental Monitoring	Show temperature, humidity and CO_2 values when available.	Admin, Energy, Operations; Proposed read-only for others
Energy Monitoring	Show room/floor energy consumption and active power.	Admin, Energy, Operations
HVAC and Lighting Monitoring	Show resource operational states.	Admin, Energy, Operations
Energy Waste Detection	Correlate occupancy with active loads and identify configured waste conditions.	Admin, Energy, Operations
Alerts	Present active waste/anomaly notifications.	Admin, Energy, Operations
Recommendations	Provide human decision-support actions linked to evidence.	Admin, Energy, Operations
Historical Analytics	Compare occupancy and energy over 24-hour, 7-day and 30-day periods.	Admin, Energy

Module	Purpose	Primary Users
Administrator Dashboard	Aggregate building load, occupancy, alerts and waste metrics.	Facility Admin
Data Simulation	Generate controlled synthetic telemetry for demos/testing.	Admin / project team
IoT/API Ingestion	Receive telemetry from REST/MQTT-connected sources where implemented.	System / authorized services

5. FUNCTIONAL REQUIREMENTS

All functional requirements use the word shall and are intended to be objectively verifiable. Acceptance criteria are expressed at requirement level where useful; Section 17 provides consolidated Given/When/Then checks.

5.1 Authentication & Access

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-01	Authentication	The system shall authenticate users before granting access to protected application features.	All users	High	Valid credentials create an authenticated session; invalid credentials are rejected.
FR-02	Logout	The system shall provide a logout action that terminates the authenticated session.	Authenticated user	High	After logout, protected pages require authentication.
FR-03	RBAC	The system shall enforce role-based access for administrative, energy, operations, and read-only user functions.	System	High	A user can access only functions permitted for the assigned role.
FR-04	Access Protection	The system shall prevent unauthenticated users from accessing protected dashboard, configuration, or data-management features.	System	High	Unauthenticated requests are redirected/rejected.

5.2 Digital Twin

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-05	Building Representation	The system shall display the current MVP building representation for SVH, Engineering Block, and Law Block.	All	High	Each mapped building can be selected and its configured floors can be accessed.
FR-06	Floor Navigation	The system shall allow users to navigate among the configured floors of a selected building.	All	High	Selecting a configured floor loads its mapped rooms.
FR-07	Room Navigation	The system shall display and preserve the supplied room IDs without modification.	All	High	All configured room IDs are rendered exactly as defined in Section 2.7.
FR-08	Room Selection	The system shall display room-specific metadata when a mapped room is selected.	All	High	Room selection opens a detail view with available metadata and current telemetry.
FR-09	Visual Status	The system shall apply configured visual status indicators to room states and alerts.	All	Medium	The displayed state corresponds to the latest valid room state.

5.3 Occupancy

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-10	Occupancy Status	The system shall display Occupied, Unoccupied, or Available status when the required data is available.	All	High	Status shown matches the current validated occupancy/schedule state.
FR-11	Occupant Count	The system shall display the current occupant count for each mapped room when a count-capable stream is available.	All	High	A valid non-negative count is shown with its timestamp.
FR-12	Occupancy History	The system shall provide historical hourly occupancy information for supported rooms.	Admin/Energy	Medium	A selected historical period returns occupancy values or an explicit incomplete-data state.

5.4 Indoor Environment

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-13	Temperature	The system shall display room temperature when a valid environmental reading is available.	Admin/Energy/ Operations	High	Current temperature and timestamp are displayed for valid telemetry.
FR-14	Humidity	The system shall display room relative humidity when a valid environmental reading is available.	Admin/Energy/ Operations	High	Current humidity and timestamp are displayed for valid telemetry.
FR-15	CO2	The system shall display room CO_2 concentration when a valid environmental reading is available.	Admin/Energy/ Operations	High	Current CO_2 value and timestamp are displayed for valid telemetry.
FR-16	Environmental History	The system shall provide historical environmental trends for supported parameters.	Admin/Energy	Medium	Selected history displays available values and identifies missing periods.

5.5 Energy & Resources

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-17	Energy Ingestion	The system shall ingest or simulate room-level and floor-level energy usage in kW/kWh.	System	High	Valid energy records are accepted and associated with the correct room/floor.
FR-18	Energy Display	The system shall display current active power and/or energy consumed for supported rooms.	Admin/Energy/ Operations	High	The latest valid value is shown with a timestamp and unit.
FR-19	HVAC State	The system shall display HVAC state as Active, Idle, or OFF when valid resource-state data is available.	Admin/Energy/ Operations	High	Displayed state equals the latest validated state.
FR-20	Lighting State	The system shall display lighting state as ON, OFF, or Dimmed when valid resource-state data is available.	Admin/Energy/ Operations	High	Displayed state equals the latest validated state.
FR-21	Resource Correlation	The system shall correlate occupancy with HVAC/lighting state	System	High	The correlation engine produces a deterministic condition

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
		and energy load for supported rooms.			result from current inputs.

5.6 Waste Detection

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-22	Waste Rule	The system shall trigger a waste condition when occupancy is zero and HVAC or lighting load exceeds the configured standby threshold continuously for the configured duration.	System	High	A qualifying condition creates one corresponding waste event after the configured duration.
FR-23	Waste Metrics	The system shall calculate or display estimated wasted energy for a detected waste event when sufficient energy data is available.	System	Medium	Estimated wasted kWh is shown or marked unavailable when required data is missing.
FR-24	Waste History	The system shall retain detected waste events for historical review.	Admin/Energy/ Operations	Medium	Resolved and active events can be retrieved according to configured retention.

5.7 Alerts

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-25	Alert Generation	The system shall generate an alert record for each qualifying energy-waste event.	System	High	Alert contains room, type, timestamp, severity/status and available wasted-energy metric.
FR-26	Alert Panel	The system shall display active alerts in a filterable dashboard panel/table.	Admin/Energy/ Operations	High	Active alerts can be filtered by supported fields.
FR-27	Alert Status	The system shall support an alert lifecycle including Active and Resolved states.	System	Medium	State transitions are persisted and visible to authorized users.

5.8 Dashboard & Analytics

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-28	Dashboard Aggregation	The system shall display total building energy rate, total occupancy percentage, active alert count, and relevant system status indicators.	Facility Admin	High	Summary values reflect the latest valid aggregated data.
FR-29	Historical Analytics	The system shall provide occupancy-versus-energy charts for 24-hour, 7-day, and 30-day periods.	Admin/Energy	High	Each period returns a chart when sufficient data exists.
FR-30	Utilization View	The system shall provide room utilization information or heatmap-style visualization for supported historical data.	Admin/Energy	Medium	Rooms are represented according to available utilization data.
FR-31	Export	The system shall provide exportable	Facility Admin	Medium	Authorized user can export a generated

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
		summaries where the export feature is enabled in the MVP.			summary in the supported format.

5.9 Data Ingestion & Simulation

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-32	Simulation Engine	The system shall provide a controllable simulator for synthetic occupancy and telemetry data.	Project/Admin	High	Simulator can start/stop and produces schema-valid records.
FR-33	API Ingestion	The system shall accept validated telemetry from supported API ingestion interfaces.	System	High	Valid payloads are accepted; invalid payloads receive structured errors.
FR-34	MQTT Integration	The system shall support MQTT-based telemetry ingestion when ESP32 sensor integration is deployed.	System	Medium	A valid MQTT message can be mapped to a configured room/stream.
FR-35	Data Validation	The system shall validate room association, timestamps, numeric ranges, and required fields before persistence.	System	High	Invalid records are rejected or quarantined with an error reason.

5.10 Recommendations

ID	Category	Requirement	Actor	Priority	Acceptance Criteria
FR-36	Recommendation Generation	The system shall generate decision-support recommendations for configured conditions such as empty-room resource usage, high occupancy, high CO_2, abnormal temperature, or high energy use.	Admin/Energy/ Operations	Medium	A qualifying condition produces a recommendation with supporting evidence.
FR-37	Recommendation Content	Each recommendation shall contain issue, room, supporting metric, suggested action, severity, timestamp, and status.	System	Medium	All required fields are present or explicitly marked unavailable.
FR-38	No Automatic Control	The system shall not automatically issue physical control commands to HVAC, lighting, breakers, or other critical infrastructure.	System	Critical	No recommendation workflow sends an actuator command to physical infrastructure.

6. DIGITAL TWIN REQUIREMENTS

The Digital Twin is the spatial layer that binds building structure to room-level telemetry and historical state. The requirements below define only the capabilities necessary for this project.

Area	Requirement
Building representation	The system shall represent the three current MVP buildings and permit selection of a building before floor navigation.
Floors	The system shall represent configured floors for each building and associate each room with exactly one configured floor record.

Area	Requirement
Rooms	The system shall represent the supplied room IDs exactly as provided in Section 2.7.
Room selection	The system shall allow a user to select a room and inspect its available metadata and latest telemetry.
Visual status	The system shall visually distinguish normal/unoccupied/occupied/alert states using the configured UI legend.
Occupancy	The system shall associate occupancy status/count with the selected room when valid data exists.
Environment	The system shall associate temperature, humidity and CO_2 values with the selected room when valid data exists.
Energy	The system shall associate active power/energy values with the selected room when valid data exists.
HVAC / lighting	The system shall display latest HVAC and lighting states where resource telemetry exists.
History	The system shall allow supported historical room data to be opened from the room detail view.

Scope Clarification: 2D MVP vs 3D Digital Twin

Source	Stated Direction
PRD	Describes an interactive 2D spatial layout as the MVP and explicitly excludes complex 3D graphics/WebGL/VR from the MVP.
Phase-I Report	Describes the proposed Digital Twin as a three-dimensional model with interactive floors/rooms and names Three.js/React Three Fiber for 3D visualization.
Current SRS handling	Both directions are preserved. The final implementation direction is not silently selected in this SRS.
Status	TBD — project team/guide confirmation required before the visual implementation is treated as final.

If the 2D MVP direction is confirmed, the 3D visualization stack is treated as a future/alternative implementation path. If the 3D direction is confirmed, Three.js/React Three Fiber becomes the visualization implementation for the Digital Twin. In either case, the room/floor data model and telemetry requirements remain applicable.

Current MVP Building Mapping

Building	Floor	Room IDs
SVH	Ground Floor	101–112
SVH	Second Floor	201–213
Engineering Block	Ground Floor	101–108
Engineering Block	First Floor	201–208
Engineering Block	Second Floor	301–308
Engineering Block	Third Floor	401–408
Law Block	Ground Floor	101–108
Law Block	First Floor	201–208
Law Block	Second Floor	301–308
Law Block	Third Floor	401–408

No additional rooms are introduced by this SRS. Physical floor naming that is not fully established by the supplied mapping remains a scope clarification item.

7. DATA REQUIREMENTS

7.1 Data Dictionary

Entity	Field	Type	Required	Validation	Description
Building	ID	Identifier	Yes	Unique; non-empty	Building identifier.
Building	Name	Text	Yes	Non-empty	Building display name.
Building	Total Floors	Integer	Yes	Non-negative	Configured floor count.
Building	Address	Text	No	Optional	Building address; exact value TBD if not available.
Building	Gross Area	Decimal	No	Non-negative	Gross area in sq ft; optional source field.
Floor	ID	Identifier	Yes	Unique	Floor identifier.
Floor	Building_ID	Identifier	Yes	Must reference Building	Parent building.
Floor	Floor Number	Text/Integer	Yes	Configured value	Display floor number/name.
Floor	Floor Map	SVG/JSON/Text	No	Valid layout format	Spatial layout representation.
Room	ID	Identifier	Yes	Unique	Room record identifier.
Room	Floor_ID	Identifier	Yes	Must reference Floor	Parent floor.
Room	Room Number	Text	Yes	Must match supplied IDs	101–112, 201–213, 201–208, 301–308, 401–408 as mapped.
Room	Room Type	Enum/Text	No	Configured value	Classroom, Lab, Office, or other configured type.
Room	Max Capacity	Integer	No	Non-negative	Maximum supported occupancy; TBD if not supplied.
Room	Standby Power Threshold	Decimal	No	Non-negative; configured	Threshold used by waste rules; exact value TBD except documented examples.
Occupancy Stream	Stream_ID	Identifier	Yes	Unique	Telemetry stream identifier.
Occupancy Stream	Room_ID	Identifier	Yes	Must reference Room	Room associated with occupancy.
Occupancy Stream	Timestamp	DateTime	Yes	Not future; valid format	Time of measurement.
Occupancy Stream	IsOccupied	Boolean	Yes	True/False	Binary presence state.
Occupancy Stream	OccupantCount	Integer	Yes if count stream	>=0	Aggregate headcount.
Energy Stream	Stream_ID	Identifier	Yes	Unique	Energy stream identifier.
Energy Stream	Room_ID	Identifier	Yes	Must reference Room	Room associated with energy data.
Energy Stream	Timestamp	DateTime	Yes	Not future; valid format	Time of measurement.
Energy Stream	ActivePower	Decimal	Yes	Non-negative	Current active power in kW.
Energy Stream	EnergyConsumed	Decimal	Yes	Non-negative	Energy consumed in kWh.
Resource State	State_ID	Identifier	Yes	Unique	Resource-state record identifier.
Resource State	Room_ID	Identifier	Yes	Must reference Room	Room associated with resource state.
Resource State	Timestamp	DateTime	Yes	Not future; valid format	Time of state.
Resource State	LightingState	Enum	Yes	ON/OFF/Dimmed	Lighting operational state.
Resource State	HVACState	Enum	Yes	Active/Idle/OFF	HVAC operational state.

Entity	Field	Type	Required	Validation	Description
Alert	Alert_ID	Identifier	Yes	Unique	Alert identifier.
Alert	Room_ID	Identifier	Yes	Must reference Room	Affected room.
Alert	Timestamp	DateTime	Yes	Valid timestamp	Alert creation time.
Alert	AlertType	Enum/Text	Yes	Configured type	Waste or other configured alert type.
Alert	Severity	Enum/Text	Yes	Configured severity	Priority/severity of the alert.
Alert	Status	Enum	Yes	Active/Resolved	Current alert lifecycle state.
Alert	EstimatedWastedkWh	Decimal	No	Non-negative	Estimated wasted energy where calculable.
Recommendation	Issue	Text	Yes	Non-empty	Detected issue summarized for human action.
Recommendation	Room	Identifier	Yes	Must reference Room	Room requiring attention.
Recommendation	Supporting Metric	Text/JSON	Yes	Must reference evidence	Metric supporting recommendation.
Recommendation	Suggested Action	Text	Yes	Non-empty	Human decision-support action.
Recommendation	Severity	Enum/Text	Yes	Configured	Recommendation severity.
Recommendation	Timestamp	DateTime	Yes	Valid timestamp	Generation time.
Recommendation	Status	Enum	Yes	Configured lifecycle	Recommendation state.
User	ID	Identifier	Yes	Unique	User identifier.
User	Role	Enum	Yes	Configured roles	Facility Admin, Energy Manager, Operations Staff, Faculty/Student.

8. BUSINESS RULES AND VALIDATIONS

Rule ID	Area	Rule	System Response
BR-01	Occupant count	Occupant count shall not be negative.	Reject invalid record and retain an error reason.
BR-02	Room association	Every telemetry record shall identify a valid configured building/floor/room relationship.	Reject or quarantine unmatched record.
BR-03	Missing telemetry	Missing telemetry shall not be treated as zero unless the source explicitly supplies zero.	Show unavailable/missing state.
BR-04	Invalid energy	Active power and energy values shall not be negative.	Reject or quarantine invalid record.
BR-05	Environmental values	Temperature, humidity and CO_2 values shall pass configured validation ranges; exact environmental thresholds are TBD.	Reject/quarantine values outside configured data-quality range.
BR-06	Duplicate telemetry	Duplicate records with the same source stream and timestamp/value identity shall not create duplicate logical state updates.	Deduplicate or flag according to ingestion policy.
BR-07	Timestamp validity	Telemetry timestamps shall use a valid format and shall not be future timestamps.	Reject invalid/future record.

Rule ID	Area	Rule	System Response
BR-08	Capacity	Occupancy shall not exceed configured room capacity when capacity is available.	Flag data-quality anomaly; do not silently clamp the value.
BR-09	Waste condition	Zero occupancy plus load above configured standby threshold for the configured continuous duration shall create a waste condition.	Generate alert after duration is satisfied; exact threshold is configured/TBD unless explicitly defined.
BR-10	No automatic control	Alerts and recommendations shall never directly actuate physical infrastructure.	No actuator command is generated.
BR-11	Privacy	Occupancy records shall represent aggregate count/binary presence only.	No individual identity field or biometric data is stored.
BR-12	Data completeness	Historical analytics shall identify incomplete data rather than presenting missing periods as measured zero.	Show missing-data indication or gap.

9. AUTHENTICATION AND AUTHORIZATION

Requirement	Specification
Login	The system shall provide authenticated login for protected roles. Credential validation shall occur before protected resources are exposed.
Logout	The system shall terminate the active session and invalidate access to protected pages/actions.
Session management	Authenticated sessions shall expire according to the configured session policy. Exact session duration is Proposed/TBD if not defined by the project implementation.
Authentication failure	The system shall show a generic authentication failure message and shall not expose whether a specific account exists.
RBAC	The system shall map each authenticated user to one configured role and enforce permissions server-side for protected operations.
Administrative features	Configuration, rule management, data management and export functions shall be protected according to assigned permissions.
Unauthorized access	Unauthenticated requests to protected resources shall be rejected or redirected to login.
Forbidden access	Authenticated users without required permissions shall receive a forbidden response and shall not receive protected data.
API authorization	Protected API operations shall validate authentication and role/permission before processing the request.

10. NON-FUNCTIONAL REQUIREMENTS

Quality	Requirement / Target	Verification
Performance	Dashboard UI shall update within 2 seconds after state-change ingestion.	Measure ingestion-to-visible-state latency under representative test load.
Performance	30-day historical chart rendering shall not exceed 1.5 seconds for the documented dataset target.	Measure client-visible render completion for 30-day dataset.
Scalability	Architecture shall support up to 500 rooms at 5-second sampling intervals.	Load/simulation test with equivalent telemetry volume.
Availability	Web application shall maintain 99.0%	Availability measurement during

Quality	Requirement / Target	Verification
	uptime during operational building hours (07:00–22:00).	operational validation period.
Reliability	Invalid, missing, duplicate, or unavailable telemetry shall not corrupt valid room state or historical records.	Fault-injection and data-quality tests.
Security	Communication protocols shall enforce TLS 1.3 where applicable to deployed network communication.	Configuration/security verification.
Security	RBAC shall protect administrative endpoints and functions.	Role-based access tests.
Privacy	The system shall not collect PII; occupancy shall be aggregate numerical/binary data.	Data model and payload inspection.
Usability	UI shall use clean responsive desktop layouts supporting 1920×1080 and 1366×768.	Visual and interaction verification at both resolutions.
Maintainability	Building, floor, room, telemetry and rules shall be separated into maintainable logical modules.	Code/design inspection.
Extensibility	The architecture shall permit additional buildings, floors, rooms and associated monitoring data without fundamental architectural changes.	Add a sample future building in a controlled test environment without changing core services.

11. SECURITY AND PRIVACY

11.1 Confirmed Requirements

- TLS 1.3 shall protect applicable network communication.
- Role-Based Access Control shall protect administrative endpoints.
- Authentication shall be required for protected operations.
- Input validation shall be applied to telemetry and API payloads.
- The system shall not collect PII, facial images, biometric identifiers, or individual identity tracking data.
- Occupancy shall be represented as aggregate numerical headcounts or binary presence.
- Recommendations and alerts shall not contain individual identity information.

11.2 Proposed Security Practices

The following practices are proposed implementation safeguards rather than additional source-of-truth project commitments:

- Store passwords using a modern adaptive password-hashing mechanism rather than plaintext.
- Use secure secret storage/environment configuration for database, MQTT, and API credentials.
- Apply least-privilege database/API service accounts.
- Return structured but non-sensitive error messages.
- Apply common web security protections such as input sanitization, secure headers, rate limiting where appropriate, and protection against injection and broken access-control patterns.
- Log security-relevant events without storing sensitive credentials or personal identity data.

12. SYSTEM ARCHITECTURE AND WORKFLOW

The architecture separates data acquisition, API services, validation/processing, persistence, Digital Twin synchronization, analytics, and the user interface. This separation supports the hybrid simulated/real-data approach described in the Phase-I report.

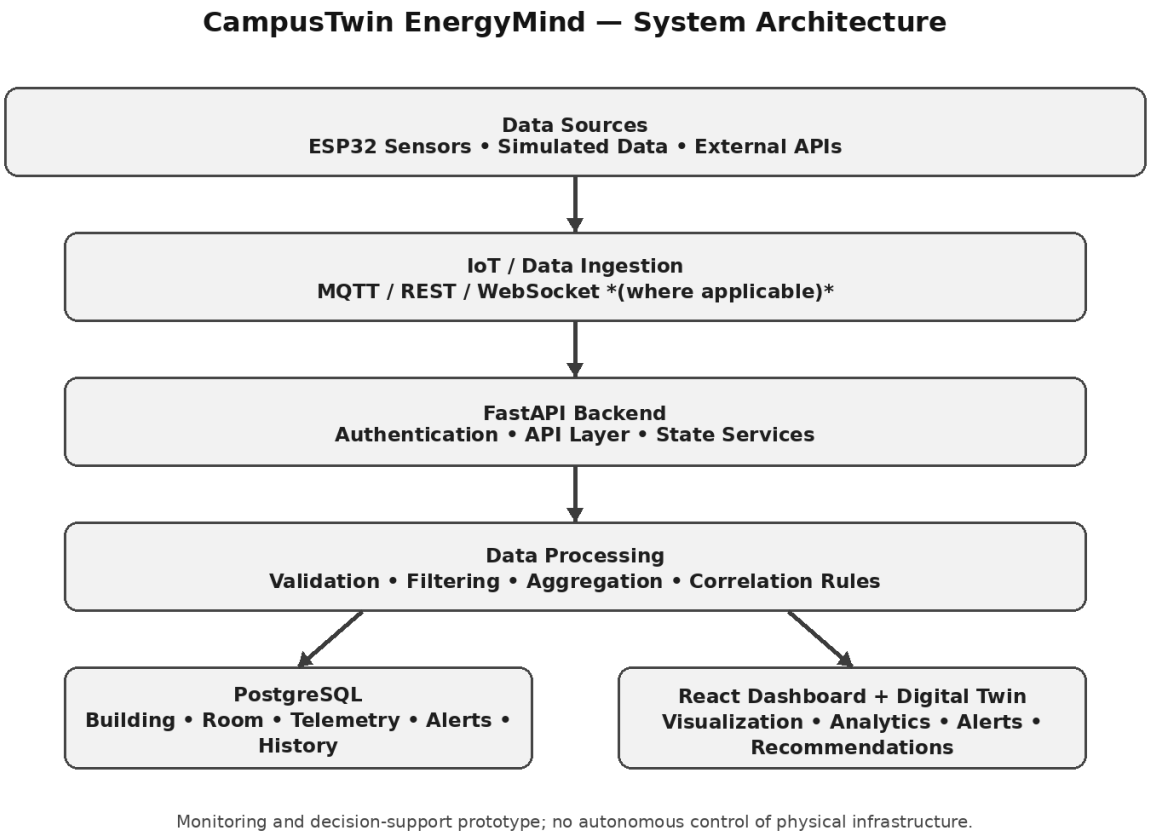


Figure 1. High-level CampusTwin EnergyMind system architecture.

12.1 Component Responsibilities

Component	Responsibility
Data Sources	Provide simulated or real occupancy, environmental, energy and resource-state data.
IoT / Ingestion	Receive data through supported REST/WebSocket interfaces and MQTT when deployed.
FastAPI Backend	Expose authenticated application/data services and mediate frontend/database interaction.
Data Processing	Validate, filter, aggregate and correlate telemetry; identify configured rule conditions.
PostgreSQL	Persist building structure, room metadata, telemetry, resource states, alerts, recommendations and history.
Digital Twin Core	Synchronize the latest valid room/building state with the spatial representation.
Dashboard / Analytics	Render current state, historical charts, alerts, recommendations and administrative summaries.

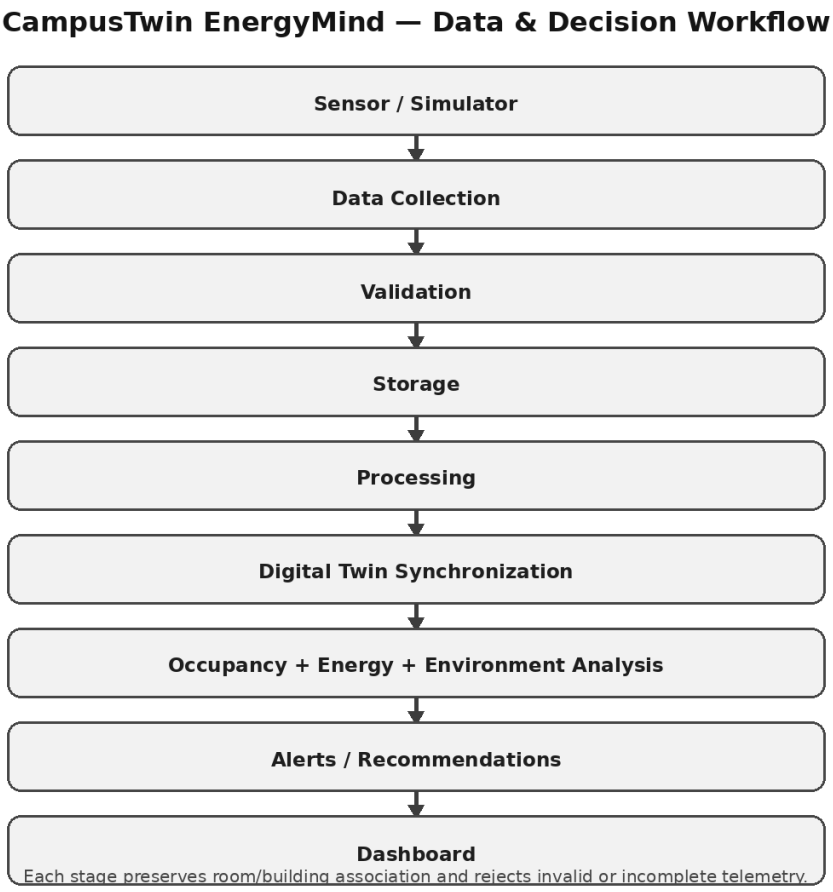


Figure 2. Telemetry-to-decision workflow.

12.2 Workflow Explanation

1. Data Collection: sensor, simulator, or API sources produce time-stamped records.
2. Validation: records are checked for required fields, room association, timestamp validity, numeric validity, and duplication.
3. Storage: validated data is persisted in PostgreSQL for current-state retrieval and historical analysis.
4. Processing: telemetry is filtered/aggregated and correlated across occupancy, energy, environmental, HVAC, and lighting data.
5. Digital Twin Synchronization: the latest valid room state is bound to the selected spatial representation.
6. Analysis: the system compares occupancy with active resource loads and historical energy/environmental patterns.
7. Alerts / Recommendations: configured conditions create human-readable decision-support outputs.
8. Dashboard: authorized users view current status, historical trends, alerts, and recommendations.

13. API AND DATA INGESTION REQUIREMENTS

Exact API endpoint names are not finalized in the source documents. The interface categories below are therefore a **Recommended API Design — Proposed**, while the behavioral requirements are derived from the project scope.

Category	Purpose	Input	Output	Auth.	Validation / Error
Authentication	Login/logout/session operations.	Credentials/session data.	Session result / status.	Required for protected actions.	Invalid credentials → structured authentication error.
Building / Floor / Room	Retrieve configured spatial structure and metadata.	Building/floor/room identifiers.	Configured records and room metadata.	Role-dependent.	Unknown ID → not-found response.
Occupancy	Ingest/retrieve current and historical occupancy.	Room ID, timestamp, occupied flag, count.	Accepted state/history.	Protected ingestion; read role-dependent.	Negative count/future timestamp → validation error.

Category	Purpose	Input	Output	Auth.	Validation / Error
Energy	Ingest/retrieve kW/kWh values.	Room ID, timestamp, power/energy.	Accepted state/history.	Protected ingestion.	Negative/invalid values → validation error.
Environment	Ingest/retrieve temperature, humidity, CO_2.	Room ID, timestamp, values.	Accepted state/history.	Protected ingestion.	Invalid range/data quality → validation error or quarantine.
Resource State	Ingest/retrieve HVAC and lighting state.	Room ID, timestamp, enum states.	Accepted state/history.	Protected ingestion.	Invalid enum → validation error.
Alerts	Retrieve/create/ resolve alert records.	Room, type, evidence, status.	Alert record/status.	Role-dependent.	Invalid lifecycle transition → error.
Recommendations	Retrieve decision-support recommendations.	Condition/evidence/ status.	Recommendation record.	Role-dependent.	Missing evidence → do not create unsupported recommendation.
Historical Analytics	Retrieve aggregated trends for 24h/7d/30d or supported ranges.	Room/building, date range.	Time-series/aggregate data.	Role-dependent.	Invalid range → validation error.
Simulation	Start/stop/configure synthetic telemetry.	Simulation configuration.	Simulation status / generated stream state.	Admin/Proposed.	Invalid configuration → validation error.

Proposed interface convention: REST-style JSON endpoints may be used for standard application operations; MQTT may be used for device-to-ingestion telemetry where ESP32 integration is deployed. Exact paths, HTTP methods, payload schemas, authentication mechanism, and error codes remain implementation decisions/TBD.

14. ERROR HANDLING AND EDGE CASES

Situation	System Response	User Message
Invalid login	Reject authentication attempt.	Invalid username or password.
Unauthorized access	Reject unauthenticated request and redirect/show login as appropriate.	Authentication required.
Forbidden role	Reject operation without exposing protected data.	You do not have permission to perform this action.
Sensor offline	Retain last known valid state with timestamp and mark telemetry unavailable/stale.	Live data unavailable; last update: <timestamp>.
Missing telemetry	Do not substitute measured zero; display unavailable/missing state.	Data unavailable for this room/metric.
Invalid telemetry	Reject/quarantine invalid record and log validation reason.	Telemetry could not be accepted.
Duplicate telemetry	Deduplicate logical duplicate or flag it without duplicating state/history.	Duplicate telemetry ignored/flagged.
Database unavailable	Return controlled service error; preserve client stability.	Service temporarily unavailable.
API unavailable	Show non-blocking error/empty state and permit retry.	Unable to load current data. Retry.
Network timeout	Abort after configured timeout and show retry state.	Request timed out.
Occupancy > capacity	Flag data-quality anomaly without silently changing the received count.	Occupancy exceeds configured capacity; verify source.
Occupancy but no energy	Show occupancy and energy as unavailable separately.	Energy data unavailable.
Energy but no occupancy	Show energy state and mark occupancy unavailable; do not infer zero occupancy.	Occupancy data unavailable.
Incomplete historical data	Render available data with visible	Historical data incomplete for selected

Situation	System Response	User Message
	gaps/coverage indicator.	period.
Session expiry	Invalidate protected state and require re-authentication.	Session expired. Please sign in again.

15. RECOMMENDATION REQUIREMENTS

The recommendation module is a decision-support component. It shall present evidence-based operational suggestions and shall not directly control physical infrastructure.

Trigger / Condition	Example Recommendation	Evidence
Empty room + HVAC ON	Inspect the room and switch HVAC off if operationally appropriate.	Occupancy = 0; HVAC state; duration; power/energy.
Empty room + Lighting ON	Check whether lighting is required and switch off if appropriate.	Occupancy = 0; lighting state; duration.
High occupancy	Review room capacity/utilization and consider operational alternatives where required.	Occupant count and configured capacity.
High CO_2	Inspect ventilation/air-quality conditions and follow facility procedure.	CO_2 reading and timestamp.
Abnormal temperature	Review HVAC/environmental conditions for the affected room.	Temperature reading and trend.
High energy consumption	Inspect resource usage and compare with occupancy/baseline context.	Active power/energy and occupancy.
Prolonged resource usage	Prioritize the room for physical inspection.	Duration and resource state.

15.1 Recommendation Record

Field	Required Content
Issue	What condition was detected.
Room	Affected room ID.
Supporting metric	Measured value(s), duration, or comparison supporting the recommendation.
Suggested action	Human-operational action; not an automatic actuator command.
Severity	Configured severity.
Timestamp	Generation time.
Status	Configured lifecycle such as New/Acknowledged/Resolved; exact lifecycle is Proposed/TBD.

The system shall never generate an instruction that directly switches a breaker, HVAC unit, lighting circuit, valve, or other critical physical infrastructure.

16. UI REQUIREMENTS

Screen	Purpose	Main Components / Information	Actions	Roles	Loading / Empty / Error
16.1 Login Page	Authenticate users.	Username/identifier, password, sign-in state, generic error.	Sign in.	All users.	Loading on submit; empty form validation; generic failure message.
16.2 Main Dashboard	Provide high-level operational view.	Building load, occupancy %, active alerts, system health, selected building/floor, alerts summary.	Filter, open room/alert, navigate modules.	Admin/Energy/ Operations; read-only subset.	Skeleton; no-data state; API error with retry.
16.3 Digital Twin /	Show spatial building	Building selector, floor	Select	All; detail role-	Loading map; no

Screen	Purpose	Main Components / Information	Actions	Roles	Loading / Empty / Error
Building View	state.	selector, room nodes, legend, status overlays.	building/floor/room.	dependent.	layout/data state; map-load error.
16.4 Floor Navigation	Move among mapped floors.	Floor list/selector and current floor label.	Select configured floor.	All.	Loading floor map; no configured floor; invalid floor error.
16.5 Room Detail Panel	Show room-specific current and recent state.	Room ID, capacity if known, occupancy, temperature, humidity, CO_2, power/energy, HVAC, lighting, timestamp, 24-hour trend where available.	Open history, inspect alert/recommendation.	All with role-dependent fields.	Loading telemetry; unavailable metrics; room-data error.
16.6 Alerts Page	Review energy-waste events.	Room, trigger time, issue type, duration, wasted kWh where available, severity, status.	Filter, acknowledge/resolve if permission exists.	Admin/Energy/ Operations.	Loading; no alerts; data-load error.
16.7 Recommendations Page	Review decision-support suggestions.	Issue, room, metric, suggested action, severity, timestamp, status.	Open/acknowledge/ resolve if configured.	Admin/Energy/ Operations.	Loading; no recommendations; error state.
16.8 Historical Analytics	Compare occupancy with energy and environment.	24h/7d/30d selectors, line/bar charts, date range, room/building filters.	Filter and inspect trends.	Admin/Energy; proposed Operations.	Loading chart; incomplete data indication; query error.

The UI shall remain usable at 1920×1080 and 1366×768 desktop resolutions. Exact visual theme, icon set, and component library are implementation choices unless separately approved.

17. ACCEPTANCE CRITERIA

AC-01	Login	**Given a registered user with valid credentials, when the user submits the login form, then the system shall establish an authenticated session and open the permitted dashboard.**
AC-02	RBAC	Given a user with a defined role, when the user opens a protected function, then access shall match the permission matrix and unauthorized functions shall be rejected.
AC-03	Building navigation	Given an authenticated user, when the user selects SVH, Engineering Block, or Law Block, then the system shall display the configured building and its mapped floors.
AC-04	Floor navigation	Given a selected building, when the user selects a configured floor, then the system shall load that floor's configured room IDs.
AC-05	Room selection	Given a mapped room, when the user selects it, then the room detail panel shall display its ID and all available current metadata/telemetry.
AC-06	Occupancy	Given valid occupancy data, when a room is opened, then the current occupancy status and count (when available) shall be displayed with a timestamp.
AC-07	Temperature	Given valid temperature telemetry, when the room detail panel is opened, then the latest temperature shall be displayed with its timestamp.
AC-08	Humidity	Given valid humidity telemetry, when the room detail panel is opened, then the latest humidity shall be displayed with its timestamp.
AC-09	CO_2	Given valid CO_2 telemetry, when the room detail panel is opened, then the latest CO_2 value shall be displayed with its timestamp.

AC-01	**Login**	**Given a registered user with valid credentials, when the user submits the login form, then the system shall establish an authenticated session and open the permitted dashboard.**
AC-10	Energy	Given valid energy telemetry, when the room detail panel is opened, then active power and/or energy consumed shall be displayed with units.
AC-11	HVAC / lighting	Given valid resource-state telemetry, when the room is opened, then HVAC and lighting states shall match the latest validated states.
AC-12	Waste detection	Given zero occupancy and a load above the configured standby threshold continuously for the configured duration, when the correlation engine evaluates the room, then an energy-waste alert shall be created.
AC-13	Alerts	Given an active alert, when an authorized user opens the alert page, then the alert shall show room, type, duration/timestamp, severity, status and available wasted kWh.
AC-14	Recommendations	Given a configured recommendation trigger, when supporting evidence is valid, then a recommendation shall contain issue, room, supporting metric, action, severity, timestamp and status.
AC-15	Historical analytics	Given sufficient historical data, when an authorized user selects 24-hour, 7-day, or 30-day analysis, then the corresponding occupancy-versus-energy visualization shall render.
AC-16	Simulation	Given a valid simulator configuration, when the simulator is started, then schema-valid synthetic telemetry shall be produced and visible through the normal ingestion path.
AC-17	Error handling	Given invalid telemetry, when it reaches the validation layer, then it shall not be persisted as valid state and the source of failure shall be identifiable.
AC-18	Privacy / control boundary	Given any alert or recommendation condition, when the system generates an output, then it shall contain no individual identity data and shall not issue physical control commands.

18. TEST CASES

TC-01	**FR-01**	**Valid login**	**Enter valid credentials for an authorized user.**	**Authenticated session and dashboard access.**
TC-02	FR-01/02	Invalid login then logout	Submit invalid credentials; then log in and log out.	Invalid login rejected; logout ends protected session.
TC-03	FR-03/04	RBAC enforcement	Attempt an administrative function using a read-only role.	Operation is forbidden; protected data is not exposed.
TC-04	FR-05/06	Building/floor navigation	Select each MVP building and configured floor.	Correct building/floor loads with configured rooms.
TC-05	FR-07	Room ID preservation	Inspect all configured room lists.	Room IDs match supplied mapping exactly; no extra rooms appear.
TC-06	FR-08	Room selection	Select a mapped room with valid telemetry.	Room detail panel opens with available metadata/current values.
TC-07	FR-10/11	Occupancy display	Inject occupied and	Status and count update

TC-01	FR-01	Valid login	Enter valid credentials for an authorized user.	Authenticated session and dashboard access.
			unoccupied room states with counts.	correctly; negative count is rejected.
TC-08	FR-13/14/15	Environment display	Inject valid temperature, humidity and CO_2 values.	Latest values and timestamps are displayed.
TC-09	FR-17/18	Energy monitoring	Inject room/floor kW and kWh records.	Energy values are stored, retrieved and displayed with units.
TC-10	FR-19/20	Resource states	Inject HVAC and lighting states.	Displayed states match validated inputs.
TC-11	FR-22/23	Waste detection	Create zero occupancy plus above-threshold load for required duration.	One qualifying waste alert is generated with available estimated kWh.
TC-12	FR-25/26/27	Alert lifecycle	Create an alert and process permitted status action.	Alert is visible and status is persisted correctly.
TC-13	FR-29	Historical analytics	Load 24h, 7d and 30d history for a room with sufficient data.	Requested charts render and show occupancy-energy relationship.
TC-14	FR-32	Simulator	Start simulator with valid configuration.	Synthetic schema-valid telemetry is generated.
TC-15	FR-33/35	Invalid ingestion	Send malformed room ID, future timestamp and negative energy payloads.	Invalid records are rejected/quarantined with structured errors.
TC-16	FR-34	MQTT ingestion	Publish a valid room telemetry message when MQTT integration is enabled.	Message maps to correct room/stream and reaches processing/storage.
TC-17	FR-36/37	Recommendation	Trigger a configured recommendation condition.	Recommendation contains required fields and supporting evidence.
TC-18	FR-38	No physical control	Trigger a waste recommendation and inspect outbound actions.	No physical actuator command is issued.
TC-19	NFR Performance	State update latency	Measure ingestion-to-dashboard update for representative changes.	Update completes within 2 seconds.
TC-20	NFR Availability/Security	TLS and availability check	Verify deployed communication and operational uptime.	Applicable traffic uses TLS 1.3; measured availability meets 99% target during defined hours.

19. REQUIREMENTS TRACEABILITY MATRIX

Requirement ID	Feature / User Story	Acceptance Criteria	Test ID
FR-01	Authentication	AC-01	TC-01
FR-03	RBAC	AC-02	TC-03
FR-05/06	Digital Twin navigation	AC-03/04	TC-04
FR-07	Room mapping	AC-04/05	TC-05
FR-08	Room detail	AC-05	TC-06
FR-10/11	Occupancy	AC-06	TC-07
FR-13/14/15	Environment	AC-07/08/09	TC-08
FR-17/18	Energy	AC-10	TC-09
FR-19/20	HVAC/Lighting	AC-11	TC-10
FR-22/23	Waste detection	AC-12	TC-11

Requirement ID	Feature / User Story	Acceptance Criteria	Test ID
FR-25/26/27	Alerts	AC-13	TC-12
FR-29	Historical analytics	AC-15	TC-13
FR-32	Simulation	AC-16	TC-14
FR-33/35	Validation	AC-17	TC-15
FR-34	MQTT ingestion	AC-16	TC-16
FR-36/37	Recommendations	AC-14	TC-17
FR-38	Control boundary	AC-18	TC-18
NFR-Performance	2-second update target	AC-02/related	TC-19
NFR-Security/Availability	TLS 1.3 and 99% uptime	Security/availability criteria	TC-20

The RTM is intentionally compact. The full functional-requirement table in Section 5 remains the authoritative detailed requirement list; the RTM links the major requirements to acceptance and test coverage.

20. ASSUMPTIONS, RISKS AND FUTURE SCOPE

20.1 Assumptions

- Simulated data may be used initially to validate the complete software architecture.
- Real sensor integration depends on hardware availability, network access, building access, and university permissions.
- Occupancy is represented as aggregate data rather than individual identity.
- Thresholds for environmental conditions and some operational rules may be configurable/TBD.
- The current SRS MVP contains three university buildings with the supplied room mapping.
- Historical and real-time data sources can share a common schema so that simulation can later be replaced or supplemented by real telemetry.

20.2 Risks

Risk	Potential Impact	Mitigation / Handling
Sensor availability	Real-time hardware validation may be delayed.	Use simulator and validate ingestion with representative synthetic data.
Data quality	Invalid/noisy readings may produce incorrect state or alerts.	Validation, filtering, deduplication and visible data-quality states.
Network connectivity	Telemetry may be delayed or unavailable.	Show stale/missing states and support retry/reconnection mechanisms.
Building access	Physical mapping or sensor deployment may require permissions.	Use available floorplan assets and simulation until access is approved.
Hardware cost	Full-room deployment may be impractical for MVP.	Use selected ESP32/sensor deployments and simulation for remaining rooms.
Incomplete real-world telemetry	Some rooms may not provide all metrics.	Display each metric independently as unavailable rather than inferring zero.

20.3 Future Scope

- Additional university buildings, floors, and rooms; the architecture shall support such additions without fundamental redesign.
- Additional occupancy, environmental and energy sensors.
- Occupancy prediction using historical building data and Python/Scikit-learn.
- Predictive maintenance and more advanced analytics, subject to project approval.
- Integration with existing Building Management Systems such as BACnet/Modbus in a future phase, with control authority governed separately.
- Campus-wide multi-building Digital Twin federation.
- Integration with university timetabling systems to compare scheduled and actual occupancy.

- Carbon/emissions tracking per space, if required in a later approved scope.

Appendix A — Requirement Baseline Notes

This SRS was derived from the uploaded PRD and Phase-I Project Report and from the project-team scope supplied with the SRS request. The PRD is the primary source for MVP product goals, functional requirements, performance targets, user stories, data entities, alerting logic, and explicit non-goals. The Phase-I report is the primary source for the project's 3D Digital Twin concept, IoT/ESP32/MQTT approach, React/Three.js/React Three Fiber, Python/FastAPI, PostgreSQL, and possible occupancy prediction using Python/Scikit-learn.

Two source-level differences are retained explicitly: (1) the PRD defines a 2D one-building MVP while the Phase-I report describes a 3D university-building Digital Twin; and (2) the current SRS request expands the MVP mapping to three IILM buildings. The SRS does not hide these differences. The three-building mapping is treated as the current project-team baseline supplied for this document, while the 2D/3D implementation direction remains TBD pending guide/team confirmation.

Appendix B — Current MVP Room Inventory

Building	Floor	Room Count	Room IDs
SVH	Ground Floor	12	101–112
SVH	Second Floor	13	201–213
Engineering Block	Ground Floor	8	101–108
Engineering Block	First Floor	8	201–208
Engineering Block	Second Floor	8	301–308
Engineering Block	Third Floor	8	401–408
Law Block	Ground Floor	8	101–108
Law Block	First Floor	8	201–208
Law Block	Second Floor	8	301–308
Law Block	Third Floor	8	401–408
Total	Current MVP	89	No additional rooms defined

The inventory contains 89 mapped rooms across the three current MVP buildings. The count is derived directly from the supplied room ranges; it does not add any unlisted rooms.

Appendix C — Key Scope Boundary

Included	Excluded / Not Finalized
Room-level occupancy, temperature, humidity, CO_2, energy, HVAC and lighting monitoring where data is available.	Autonomous physical control of HVAC, lighting, breakers or critical infrastructure.
Simulated telemetry and future real ESP32/MQTT integration.	Facial recognition, biometrics, individual identity tracking.
Alerts and recommendations as decision support.	Native mobile apps and VR/AR walkthroughs.
24-hour, 7-day and 30-day historical analytics.	Unapproved advanced predictive maintenance / forecasting.
Extensible building/floor/room architecture.	Final 2D vs 3D visualization direction — TBD.